
3a	base units: kg m s^{-2} × m = kg m^2 s^{-2}	A1
3bi	distance of COG from P (= GP) = 17 cos 45° – 4.0 = 8.02 cm (or using Pythagoras Theorem: $\sqrt{144.5} - 4.0 = 8.02$) moment = 0.15 × 8.02 × 10^{-2}	C1 A1

	$= 1.2 \times 10^{-2}$ N m	
3bii	(line of action of) weight acts through pivot/P **or** distance between (line of action of) weight and pivot/P is zero (so) weight does not have a moment about pivot/P	M1 A1
3ci	upthrust = 6.20 – 5.60 = 0.60 N	C1 A1
3cii	$\Delta p = \Delta F / A$ $= 0.60 / 1.2 \times 10^{-3}$ = 500 Pa	 C1 A1
3ciii	upthrust increases when density increases and since upthrust + force on spring = weight of cylinder so extension decreases	M1 A1

4a	The heat input is used to break intermolecular bonds between water molecules/	B1